

4D Embroidery: Implementing Parametric Structures in Textiles for Sculptural Embroidery

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[th]



Figure 1: Fabrication Workflow

ABSTRACT

4D Embroidery, a method to create sculptural 3D structures from a flat sheet of fabric by parameterizing an open-lace embroidery framework. By adding a fourth dimension to a mesh-like texture in textiles, lace has the potential to exhibit a wide range of transformation behavior for actuation. However, digital embroidery is not widely adopted as a technique central to fabricating 4D structures. This study presents a design exploration of the methodologies for implementing parametric structures in self-assembling textiles. We propose a digital fabrication process via embroidering active and passive material, accompanied by a computational design tool to achieve a target shape-change. We reflect on the design process and present future directions for research.

CCS CONCEPTS

• Applied computing → Computer-aided manufacturing.

KEYWORDS

Fabrication; Embroidery; Material Programming; Shape-changing; Smart Materials

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1 INTRODUCTION

Employing additive manufacturing processes to generate self-assembling textiles has recently gained traction among product, fashion, automotive and architecture industries [3]. The appeal can be attributed to the prolific use of 3D printing programmable matters, which can be described as four-dimensional (4D) printing. The fourth dimension is the temporal form or the time when shape-change occurs in addition to the three-dimensional shape [5]. Many research accounts have investigated 4D transformation on textiles by studying the smart behavior of 3D printing two-dimensional patterns of elastic fibers on pre-stretched textiles to create self-morphing objects. However, this approach only treats textiles as a substrate by adding extraneous accessories to garments, and this way could increase cost, complexity, failure, and bulkiness. On the other hand, current general fabrication devices have received much less attention. This also applies to industrial-grade textile machines enabling scalability and mass production. In this work-in-progress (WIP) paper we present 4D Embroidery, a novel approach to programming textiles using the parameterization tool of open-lace sculptural embroidery which has a two-layer structure: Heat activated shrinking layer and a passive normal layer. The goal is four-fold: (1) to develop a fabrication method; (2) to define the design space; (3) to develop a design tool and fabrication workflow; (4) to showcase multiple application scenarios. Furthermore, 4D Embroidery adaptation to wearable systems will help cut down on time, manufacturing and shipping costs.

2 RELATED WORK

There is not much research done that explores self-assembling and self-shaping structures in textiles using textile machines. The closest related work is Active Textiles which tailors custom-fit clothing

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responsive to heat and moisture using an industrial knitting machine [4]. Utilizing stitch-level control programming into the fabric, shrinking yarns are selectively placed in combination with passive threads. Similarly, Albaugh et al's work employed new techniques for (motor-controlled) yarn inlay actuation embedded directly in the knitting process [1] by shaping basic topologies, combining anisotropy knit or purl texture. Consequently, digital embroidery has the potential to exhibit interesting transformation behavior in textiles. A visual artist exhibited her work on sculptural embroidery by molding lace patterned embroidery between two sandwiching bowls into a dome shape [6]. Her work entails delicate embroidery artwork exhibiting the aesthetic quality and complexity of mesh structures. This existing technique for lace embroidery has never been examined before for the application of self-shaping and/or self-assembling structures.

3 FABRICATION METHOD AND MATERIALS

The current fabrication workflow for this study (Figure 1) requires two layers or threads to produce an embroidery of approx. 75mm x 15mm x 2mm using Parie-EMM1901 manufactured by Brother Industries Ltd.¹, and SI-SHU-PRO-11 embroidery software². As the active

material, we used a special polyester (PE) thread (Chizimi³) that undergoes 30% shrinkage when exposed to heat. In contrast, a generic PE thread is used for the underlay and frame as a restrictive layer to control the bending direction. As a result, the embroidery will bend in the direction perpendicular to where the sewn Chizimi is facing. Through both digital and material experiments, we examined the relationship between the number of running Chizimi stitches or density, from one to five lines respectively for each sample, and their bend angle once submerged in 63°C of hot water. All five samples shown here have a set of fixed parameters that first need configuring right before embroidering, namely: (Step 1) 6 lines/mm tatami stitch, 4mm sewing pitch, 0% angle deviation, and 2mm running stitch length for the underlay or base; (Step 2) 2mm of running stitch for chizimi upper thread to be sewn on top of the base layer;

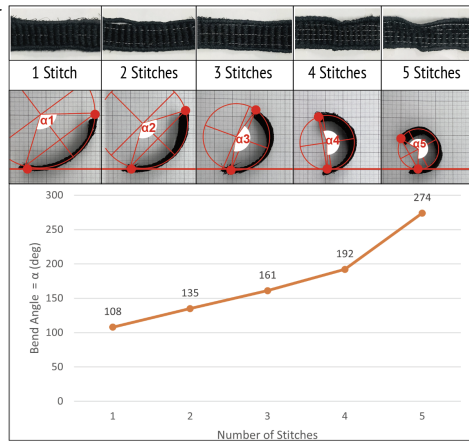


Figure 2: The relationship between the number of shrinking thread or Chizimi lines and bend angle

(Step 3) The satin stitch frame is 3.5mm wide with a density of 4 lines/mm. As a result, the more Chizimi lines that are stitched, the more the sample will curve. As shown in Figure 2, the maximum curvature out of all samples almost resulted in a full circle shape and the graph also showed 82% angle difference which is the largest change among the samples, whereas the former samples produced a more linear changes in curvature.

Also, multiple open-pattern samples were produced using 2-ply stainless thin conductive thread from Adafruit⁴ as the bobbin to demonstrate the interactive capabilities of 4D Embroidery (Figure 3). The open patterns were used to produce 4D embroidered push buttons with different levels of actuation force, from linear to tactile, and Chizimi layers were positioned accordingly on both surfaces to build the structure. Then, conductive threads were used as the bobbin for the button to hit the embroidered contact switches.

4 DISCUSSION AND FUTURE WORK

Given the inherent material properties of embroidery, it has the potential to be programmed to exhibit various and possibly contrasting material qualities depending on the requirements when combined with 4D Embroidery shape transformation and interactive capabilities. Hence, a plethora of novel material expressions

may be explored in future research. The next step is to conduct scientific experiments with other geometric parameters such as stitching patterns and directions to find the relationship with bending orientation such as twisting behavior. Also, to investigate different sensing modalities that can be produced from 4D Embroideries such as pressure and touch to detect performative and/or affective gestures. Then ultimately, the design process will be streamlined into a single tool using the experimental data from this study. This will be done with the help of the pyembroideryGH library to directly manipulate embroidery format and simulate the shape-change [2]. Later, adding patterns with increasing complexity to the equation, conducting user studies and defining the design space for 4D Embroidery. This will consequently provide new expressive interaction capabilities for wearables that researchers, designers, hobbyists and DIY enthusiasts can personalize based on their individual needs.

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³ <https://www.it-ozeki.com/> (accessed 2022-08-05)



Figure 3: 4D Embroidery transformation from flat to arched state (top); And an audience playing a Pong game with the embroidered push buttons (bottom)

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